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Genetic Programming-Based Evolutionary Strategy Generation for Complex Optimization

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Abstract—Designing effective update strategies is vital to the success of metaheuristic algorithms. Traditional methods often depend on manually designed rules and empirical tuning, which restrict their adaptability and scalability across different optimization problems. To overcome this limitation, a novel framework named GP-MAs is proposed, which integrates Genetic Programming (GP) into metaheuristics to automatically evolve and optimize their update equations. In this framework, GP dynamically constructs learning rules within the search process, allowing the algorithm to adapt to varying problem landscapes. The framework is implemented on the Growth Optimizer (GO), forming a hybrid variant called GPbasedGO. Experimental evaluations on the CEC2022 benchmark suite demonstrate that GPbasedGO achieves superior convergence speed, robustness, and generalization ability compared with several state-of-the-art algorithms. The proposed GP-MAs framework offers a flexible and automated paradigm for metaheuristic design, enabling the generation of adaptive update strategies suitable for complex optimization tasks and real-world applications.

Index Terms—Metaheuristic optimization, Genetic Programming, Adaptive update strategy

I. INTRODUCTION

With the rapid advancement of artificial intelligence (AI), applications in healthcare, transportation, and manufacturing have grown substantially. However, challenges such as technical bottlenecks, data heterogeneity, and scenario-specific constraints persist. Real-world problems often exhibit non-linear, non-differentiable, and complex characteristics. Metaheuristic algorithms (MAs) have emerged as powerful tools for global optimization, providing flexible, structure-independent solutions [1], [2]. Unlike traditional optimization, MAs explore complex search spaces probabilistically, often simulating natural, biological, or physical processes, and exploit problem-specific knowledge to find high-quality solutions. Recent applications include wireless sensor networks (WSNs) [3], [4], cloud resource allocation [5], communication security [6], intelligent manufacturing [7], machine learning [8], and AI tasks [9], [10].

MAs are taxonomically classified based on inspiration mechanisms [11], [12], as summarized in Fig. 1.

Classification: 1) *Evolutionary-based:* Genetic Algorithm

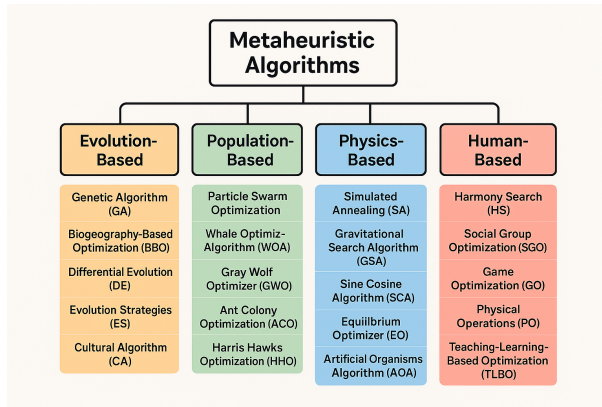


Fig. 1: A brief classification of metaheuristic algorithms.

(GA) [13], Differential Evolution (DE) [14], Evolution Strategy (ES) [15], Biogeography-Based Optimization (BBO) [16], Immunization Algorithm (IA) [17]; 2) *Swarm intelligence*: PSO [18], ACO [19], ABC [20], GWO [21], WOA [22], MGO [23]; 3) *Physics-based*: Simulated Annealing (SA) [24], Gravitational Search Algorithm (GSA) [25], Sine Cosine Algorithm (SCA) [26], Multi-Verse Optimization (MVO) [27]; 4) *Human behavior-inspired*: TLBO [28], Harmony Search (HS) [29], Social Group Optimization (SGO) [30], Special Forces Algorithm (SFA) [31].

Research in MAs follows three main directions: (i) proposing new algorithms, e.g., Gannet Optimization Algorithm (GOA) [32] and Status-based Optimization (SBO) [33]; (ii) hybridization of algorithms, e.g., IMPAEO [34] and HADEFP [35]; (iii) enhancing performance through new strategies or parameter tuning [36].

Despite their diversity, MAs rely heavily on manually designed update rules, often requiring empirical tuning. These dependencies introduce several critical limitations:

- **Empirical dependence**: strategies are often problem-specific, leading to performance fluctuations across different tasks;
- **Lack of adaptability**: static rules cannot dynamically adjust to the changing landscapes during the optimization process;
- **High resource consumption**: redesigning or hybridizing formulas requires substantial expert effort and computational cost.

To address these limitations, Meta-Black-Box Optimization (MetaBBO) has recently emerged as a promising direction, leveraging meta-learning to automate the design and adaptation of BBO algorithms with minimal expert intervention [37], [38]. MetaBBO typically adopts a bi-level framework, where a meta-level policy observes the optimization dynamics of a low-level BBO process and adaptively determines algorithmic designs based on performance feedback over a problem distribution.

Existing MetaBBO studies can be categorized into algorithm selection [39], algorithm configuration [40], solution manipulation [41], and algorithm generation [42], while var-

ious learning paradigms, including reinforcement learning [43], supervised learning [44], neuroevolution [45], and LLM-based in-context learning [46], have been explored. Moreover, MetaBBO has been instantiated across diverse optimization scenarios, such as single-objective, multi-objective, multi-modal, large-scale, and multi-task optimization [47].

Despite the rapid progress of Meta-Black-Box Optimization (MetaBBO) [48], most existing approaches predominantly rely on neural-network-based meta policies, particularly reinforcement learning [49], to achieve adaptive algorithm design through online state-action decision making. Although such policy-learning-based frameworks have demonstrated strong performance, they typically require carefully designed state representations, reward functions, and extensive training interactions, which may increase algorithmic complexity and reduce interpretability. Moreover, the reliance on online policy inference introduces additional computational overhead during optimization.

In contrast to policy-learning-based MetaBBO frameworks, this study proposes a Genetic Programming-based Metaheuristic framework, termed GP-MAs, which focuses on automatically discovering explicit update rules at the algorithmic level. Instead of learning an implicit control policy, GP-MAs evolves update formulas offline using symbolic genetic programming and embeds the resulting expressions directly into the low-level optimizer. By replacing manually crafted update rules with evolved symbolic expressions, the proposed framework significantly reduces reliance on expert knowledge while maintaining adaptability across diverse optimization landscapes and avoiding additional online decision-making costs.

To instantiate the proposed GP-MAs framework, the Growth Optimizer (GO) [50], inspired by human learning and reflection behaviors, is adopted as a representative low-level metaheuristic algorithm. By integrating GP-MAs into the learning phase of GO, a hybrid variant named GPbasedGO is developed, which is capable of automatically discovering effective update strategies and achieving a better balance between exploration and convergence. Extensive experiments conducted on the CEC2022 benchmark suite demonstrate that GPbasedGO achieves improved optimization accuracy and robustness compared with several state-of-the-art algorithms.

The remainder of the paper is structured as follows: Section II reviews GO and GP methodologies; Section III introduces the GP-MAs framework and GPbasedGO; Section IV presents experimental results; Section V concludes and discusses future directions.

II. RELATED WORK

This section systematically reviews the theoretical foundations and techniques closely related to this research, focusing primarily on the Growth Optimizer (GO) and Genetic Programming (GP). These methodologies provide the theoretical basis for the proposed framework and experimental studies.

A. Growth Optimizer

The GO algorithm is an emerging intelligent optimization technique that simulates the human processes of “learning,

reflection, and self-improvement.” It is characterized by strong global search capability and adaptability. This section outlines its fundamental structure, comprising initialization, the learning mechanism, and the reflection phase.

1) **Initialization:** The initial population (Generation 0) consists of N_p individuals. For the i -th individual $X_i(0)$, the j -th component $x_{i,j}(0)$ is initialized within the search space boundaries $[x_{i,j}^{\min}, x_{i,j}^{\max}]$, where D denotes the dimensionality of the problem. The initialization typically employs a uniform distribution:

$$x_{i,j}(0) = x_{i,j}^{\min} + \text{rand}(0, 1) \cdot (x_{i,j}^{\max} - x_{i,j}^{\min}) \quad (1)$$

2) **Learning Phase:** Individuals improve their positions by learning from the “gaps” between themselves and others. Four specific types of gaps are defined to guide this process:

$$\begin{cases} \text{Gap}_1 = \vec{x}_{\text{best}} - \vec{x}_{\text{better}} \\ \text{Gap}_2 = \vec{x}_{\text{best}} - \vec{x}_{\text{worse}} \\ \text{Gap}_3 = \vec{x}_{\text{better}} - \vec{x}_{\text{worse}} \\ \text{Gap}_4 = \vec{x}_{L1} - \vec{x}_{L2} \end{cases} \quad (2)$$

The learning factor (LF_k) quantifies the influence of each gap:

$$LF_k = \frac{\|\text{Gap}_k\|}{\sum_{k=1}^4 \|\text{Gap}_k\|}, \quad k = 1, 2, 3, 4 \quad (3)$$

The self-perception factor (SF) evaluates the state of an individual:

$$SF = \frac{GR}{GR_{\max}} \quad (4)$$

Based on these factors, the knowledge acquisition vector from each gap is calculated as:

$$\vec{K}A_k = SF \cdot LF_k \cdot \vec{Gap}_k, \quad k = 1, 2, 3, 4 \quad (5)$$

Consequently, the position update is performed as follows:

$$\vec{x}_i^{\text{it}+1} = \vec{x}_i^{\text{it}} + \sum_{k=1}^4 \vec{K}A_k \quad (6)$$

To ensure genuine improvement, an update verification step is applied:

$$\vec{x}_i^{\text{it}+1} = \begin{cases} \vec{x}_i^{\text{it}+1} & \text{if } f(\vec{x}_i^{\text{it}+1}) < f(\vec{x}_i^{\text{it}}) \\ \vec{x}_i^{\text{it}} & r_1 < P_2 \\ \vec{x}_i^{\text{it}} & \text{else} \end{cases} \quad \text{otherwise} \quad (7)$$

3) **Reflection Phase:** The reflection phase allows individuals to escape local optima by updating dimensions stochastically based on reflection probability:

$$\vec{x}_{i,j}^{\text{it}+1} = \begin{cases} x_{i,j}^{\min} + r_4 \cdot (x_{i,j}^{\max} - x_{i,j}^{\min}) & r_3 < AF \\ \vec{x}_{i,j}^{\text{it}} + r_5 \cdot (R_j - \vec{x}_{i,j}^{\text{it}}) & r_2 < P_3 \\ \vec{x}_{i,j}^{\text{it}} & \text{otherwise} \end{cases} \quad (8)$$

$$AF = 0.01 + 0.99 \cdot \left(1 - \frac{FEs}{MaxFEs}\right) \quad (9)$$

B. Genetic Programming

Genetic Programming (GP) is an evolutionary computation technique that automatically generates program structures to solve problems. Unlike traditional optimization that optimizes numerical parameters, GP evolves the structure of the solution itself. Individuals are represented as syntax trees composed of function nodes $F = \{f_1, \dots, f_n\}$ (inner nodes) and terminal nodes $T = \{t_1, \dots, t_m\}$ (leaf nodes).

Common initialization methods include the **Full Method**, where functions fill all non-leaf nodes and terminals occupy leaves, and the **Grow Method**, where nodes are randomly assigned. The **Ramped Half-and-Half** method combines both to ensure structural diversity in the initial population.

The evolutionary process in GP relies on three fundamental operations:

(1) **Selection:** This operator selects superior individuals for reproduction based on fitness. Common strategies include Roulette Wheel, Tournament, and Rank selection.

(2) **Crossover:** This operator promotes information exchange by swapping randomly selected subtrees between two parent individuals to produce offspring, as illustrated in Fig. 2.

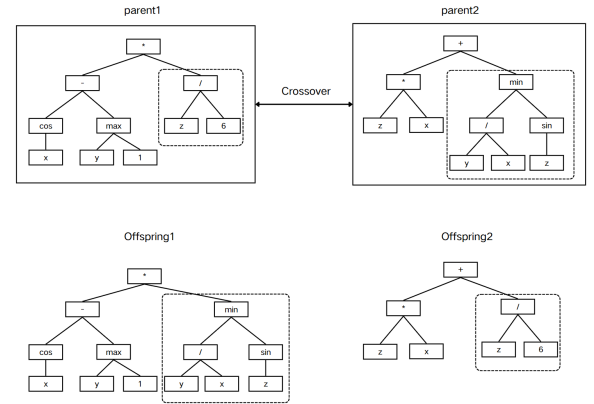


Fig. 2: Diagram of the crossover operation.

(3) **Mutation:** This operator introduces structural diversity to prevent premature convergence. It typically involves replacing a subtree with a randomly generated one (Subtree Mutation) or randomly altering a specific node (Point Mutation), as shown in Fig. 3.

III. GP-MAS FRAMEWORK AND GO ALGORITHM BASED ON THE GP-MAS FRAMEWORK

This section introduces the GP-MAs framework, outlining its architecture, parameter design, and its specific instantiation in the Growth Optimizer (GPbasedGO).

A. Overall Structure

As illustrated in Fig. 4, the GP-MAs-GO framework operates in two distinct phases. **Phase 1 (Offline Evolution):** A GP

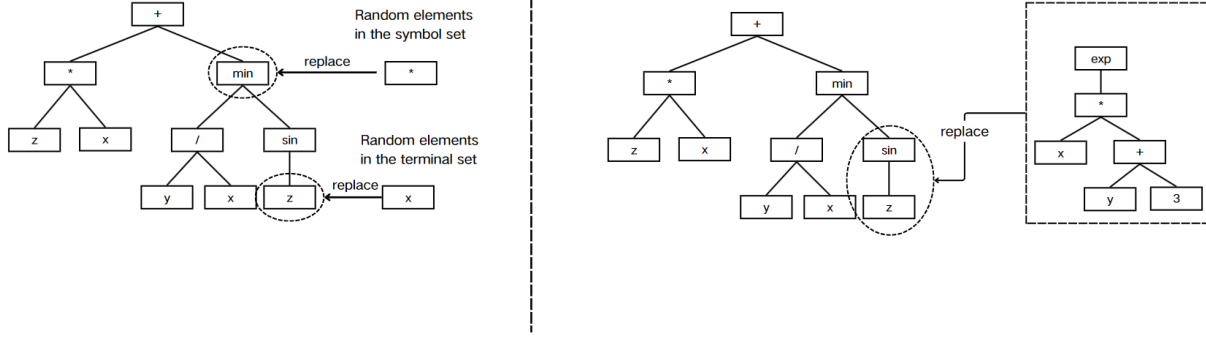


Fig. 3: Diagram of the mutation operation.

population encoding candidate update formulas is initialized and evolved. Individuals are evaluated by embedding them into the learning phase of the Growth Optimizer (GO) and measuring performance on selected benchmark functions. Through selection, crossover, and mutation, the framework iteratively refines these strategies. **Phase 2 (Online Application):** The optimal evolved formula replaces the original manual update rule in GO, yielding GPbasedGO. Since evolution occurs offline, the resulting strategy is static during deployment, reducing manual design effort while ensuring adaptability across varying landscapes.

B. The GP-MAs Framework

The GP-MAs framework utilizes Genetic Programming to automatically construct mathematical update strategies. Table I summarizes the parameter settings.

TABLE I: Parameter settings of the GP-EAs framework

Parameter	Value
Population size (PS)	100
Initialization method	Ramped Half-and-Half
Maximum iterations (MAXIT)	200
Tree depth (MAXD / MIND)	7 / 9
Elitism rate	20%
Crossover / Mutation probability	0.8 / 0.2

The workflow, detailed in Algorithm 1, begins with the initialization of a structurally diverse population. Each individual represents a candidate update rule in tree format. During evolution, individuals undergo variation (crossover and mutation) and selection (roulette-wheel). To ensure robustness, a customized fitness function F aggregates performance across four representative CEC2022 functions: F1 (unimodal), F3 (basic multimodal), F7 (hybrid), and F10 (composition). The fitness is computed as:

$$\text{Fitness} = \sum_{j \in \{1,3,7,10\}} \frac{\text{fit}_j}{\text{fit}_j^*} \quad (10)$$

where fit_j is the mean result over five runs and fit_j^* is the global optimum. This design promotes the evolution of generalized strategies capable of handling diverse optimization challenges.

C. The GP-GO Algorithm

GPbasedGO is developed by integrating the GP-evolved learning rule into the GO algorithm. The evaluation is performed on 10-dimensional instances of the training set (F1, F3, F7, F10). This low-dimensional setting is intentionally chosen to test the scalability of the strategy to higher dimensions (e.g., 20-D) in subsequent validation. The training set covers the full spectrum of CEC2022 landscape characteristics, forcing the evolved formula to balance exploration and exploitation and minimizing the risk of overfitting.

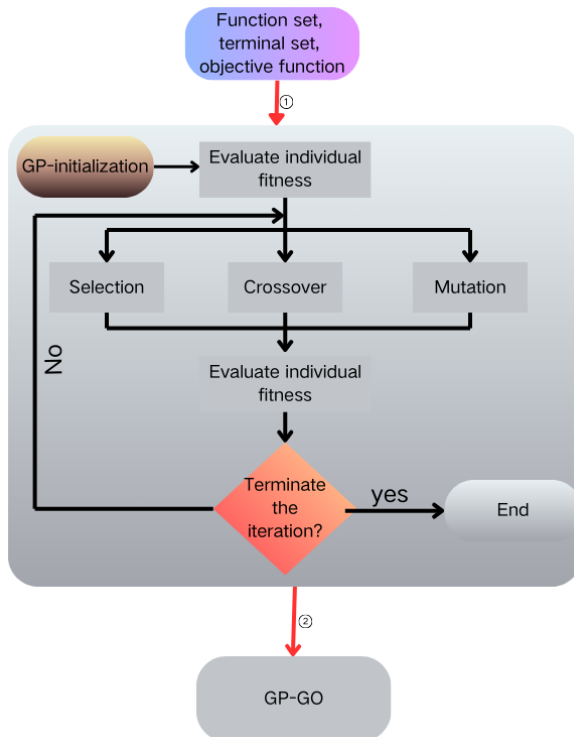


Fig. 4: System framework diagram.

Algorithm 1 GP-MAs: Genetic Programming-based Metaheuristic Automation

```

1: Input:  $PS$ ,  $MAXD$ ,  $MAXIT$ , Evaluation function  $F$ ,
   Function/Terminal Sets
2: Output: Optimal evolved update rule
3: Initialization: Generate  $POP^0$  (50% depth-first, 50%
   breadth-first)
4: Parse individuals  $\rightarrow$  executable expressions
5: Evaluate fitness using  $F$ 
6:  $iter \leftarrow 1$ 
7: while  $iter \leq MAXIT$  do
8:   Preserve top 20% (elitism); Select parents via hybrid
   selection
9:   Apply Crossover (subtree swapping) and Mutation
   (node/subtree)
10:  Evaluate valid offspring using  $F$ ; Fill  $POP^{iter+1}$ 
11:   $iter \leftarrow iter + 1$ 
12: end while
13: Return: Best individual

```

The GP performance relies on the terminal set $\mathcal{T}$ and function set $\mathcal{F}$:

$$\begin{cases} \mathcal{T} = \{Gap_1, Gap_2, Gap_3, Gap_4, SF\} \\ \mathcal{F} = \{+, -, \times, \div, \text{norm}, \sin, \cos, \tan\} \end{cases}$$

$\mathcal{T}$ captures social hierarchy information (Gap_{1-4}) and adaptive growth resistance (SF). $\mathcal{F}$ introduces non-linear perturbations via trigonometric functions and norms to maintain diversity.

The final evolved update rule is expressed as:

$$\text{Update} = \frac{SF \cdot (\text{Term}_1 + \text{Term}_2)}{\text{Denominator}_1 + \text{Denominator}_2} \quad (11)$$

where the components are defined as:

$$\begin{aligned} \text{Term}_1 &= \|Gap_1\| \cdot Gap_1 + \|Gap_2\| \cdot Gap_2 \\ \text{Term}_2 &= \|Gap_3\| \cdot Gap_3 + \|SF^2\| \cdot Gap_4 \\ \text{Denominator}_1 &= \|\cos(\tan(\cos(\|Gap_1\|)))\| + \|Gap_2\| \\ \text{Denominator}_2 &= \|Gap_3\| + \|\sin(\|\cos(Gap_3 - Gap_2)\|)\| \end{aligned} \quad (12)$$

Compared to the manual formulation, this evolved equation features higher structural complexity and information integration. It leverages non-linear operators and higher-order gap interactions (e.g., differences between gaps) to dynamically adjust search step sizes. This autonomous design effectively balances convergence speed with population diversity, offering superior adaptability to complex problems without human design bias.

IV. EXPERIMENTAL RESULTS

This section evaluates the proposed GPbasedGO using the CEC2022 benchmark suite, covering unimodal, multimodal, hybrid, and composition functions (F1–F12).

A. Experimental Setup

A comprehensive comparative study is conducted against fifteen algorithms, comprising classical methods (PSO, GA, ASO, BOA, SA, CS, GWO, DE, CRO) and recent state-of-the-art metaheuristics (GO, WOO [51], EMWPA [52], FNO [53], CHO [54], LEE [55]). To ensure fairness, all algorithms utilize identical population sizes and fixed fitness evaluations (FES). Each experiment involves 30 independent runs to ensure statistical reliability, assessed via mean, standard deviation, and the Wilcoxon rank-sum test.

B. Analysis of Experimental Results

1) *Analysis of 10-dimensional Results:* Tables II, III, and IV present the 10-dimensional comparison. GPbasedGO demonstrates superior overall performance, achieving statistically significant improvements on a wide range of functions (e.g., F1, F4, F6, F11). Compared to classical algorithms, it exhibits faster convergence and higher precision. Against recent high-performing methods like CHO and EMWPA, GPbasedGO maintains a competitive edge, particularly on complex multimodal landscapes where it effectively avoids local optima. While algorithms such as FNO and LEE show strength on specific functions, GPbasedGO offers a more balanced performance profile across the entire suite.

Fig. 5 illustrates the convergence curves for representative functions. GPbasedGO (red curve) shows a continuous downward trajectory, effectively balancing exploitation and exploration. In contrast, competitors like CS and CRO tend to stagnate early on complex functions (e.g., F2, F12), whereas GPbasedGO avoids premature convergence to achieve high-quality solutions.

2) *Analysis of 20-dimensional Results:* Tables V, VI, and VII detail the 20-dimensional results. Notably, the GPbasedGO update strategy was evolved using only 10-dimensional training data. Its superior performance on 20-dimensional test functions demonstrates strong dimensionality transferability, indicating the evolved formula captures scalable optimization mechanics rather than overfitting to the low-dimensional training set.

GPbasedGO achieves the best or near-best mean values on the majority of functions. It significantly outperforms classical algorithms in stability and remains highly competitive against recent metaheuristics (e.g., WOO, FNO). As shown in the convergence curves (Fig. 6), the transition to 20 dimensions causes many algorithms to stagnate due to the ‘‘curse of dimensionality.’’ However, GPbasedGO maintains a persistent downward trend. This resilience stems from the intrinsic scalability of the GP-evolved formula, which utilizes non-linear operators to generate diverse step sizes, preventing diversity loss and ensuring robust performance in higher-dimensional spaces.

V. CONCLUSION

This paper introduces a novel computational framework, GP-MAs, which harnesses the power of Genetic Programming (GP) to autonomously design and optimize update strategies

TABLE II: 10-Dimensional comparison results(Part 1).

Func_Num	Mean_PSO	R	Mean_GA	R	Mean_DE	R	Mean_GWO	R	Mean_GO	R	Mean_GPbasedGO
1	8828.213	+	7178.107	+	4653.567	+	5883.246	+	2070.409	+	761.238
2	481.4768	+	891.4384	+	421.6076	+	443.5214	+	408.641	+	403.9103
3	622.0307	+	641.7936	+	608.4225	+	602.7519	+	601.6679	=	601.4074
4	856.9073	+	841.992	+	850.7588	+	822.5716	=	838.2599	+	821.5802
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6	4344257	+	56128305	+	17187.74	+	21769.89	+	28638.76	+	4487.426
7	2070.12	+	2078.786	+	2052.423	+	2049.472	+	2035.152	+	2023.685
8	2237.394	+	2248.042	+	2231.638	+	2263.761	+	2228.674	+	2224.31
9	2587.122	+	2704.782	+	2506.232	=	2593.41	+	2529.35	+	2501.057
10	2624.657	+	2547.583	=	2546.931	=	2591.171	=	2547.91	=	2557.738
11	2995.133	+	3080.935	+	2828.305	+	2828.417	=	2689.166	=	2749.009
12	2875.162	+	3001.373	+	2862.248	+	2869.924	+	2863.979	+	2860.399
Win		12		11		11		8		9	

TABLE III: 10-Dimensional comparison results(Part 2).

Func_Num	Mean_ASO	R	Mean_BOA	R	Mean_SA	R	Mean_CS	R	Mean_CRO	R	Mean_GPbasedGO
1	11718.16	+	14424.11	+	10635.12	+	20780.83	+	2331248	+	761.238
2	489.1466	+	2221.291	+	526.9067	+	1651.528	+	4592.578	+	403.9103
3	620.5264	+	668.0617	+	649.047	+	668.4357	+	699.9431	+	601.4074
4	838.7514	+	875.0021	+	874.2788	+	890.413	+	928.6312	+	821.5802
5	1041.59	+	1787.522	+	1528.244	+	2680.351	+	4934.607	+	914.3811
6	1004955	+	2.58E+08	+	4323.045	=	1.56E+08	+	1.62E+09	+	4487.426
7	2087.336	+	2123.092	+	2106.302	+	2144.093	+	2274.834	+	2023.685
8	2240.496	+	2317.531	+	2249.618	+	2313.266	+	3270.592	+	2224.31
9	2652.427	+	2823.84	+	2724.674	+	2819.166	+	3176.525	+	2501.057
10	2517.251	=	2859.451	+	2756.895	+	2738.914	+	4144.509	+	2557.738
11	2796.109	+	3811.853	+	3111.818	+	3743.182	+	5064.099	+	2749.009
12	2927.294	+	3085.632	+	2963.107	+	2977.527	+	3238.392	+	2860.399
Win		11		12		11		12		12	

TABLE IV: 10-Dimensional comparison results (Part 3).

Func_Num	Mean_WOO	R	Mean_EMWPA	R	Mean_FNO	R	Mean_CHO	R	Mean_LEE	R	Mean_GPbasedGO
1	6536.274	+	5690.298	+	2317.047	+	11615.56	+	3841.837	+	761.238
2	426.5368	+	431.3476	+	403.4368	=	689.9362	+	415.7015	+	403.9103
3	614.2211	+	631.5148	+	601.7645	=	623.9042	+	601.866	+	601.4074
4	842.7871	+	832.3918	+	819.6097	=	828.5045	=	821.1671	=	821.5802
5	1040.472	+	1148.165	+	905.6369	=	1099.184	+	903.5064	=	914.3811
6	291645.1	+	3238.29	=	3255.117	=	28586566	=	20309.67	+	4487.426
7	2052.795	+	2074.466	+	2025.991	=	2064.781	+	2036.688	+	2023.685
8	2231.118	+	2258.296	+	2228.317	+	2234.951	+	2228.119	+	2224.31
9	2547.921	+	2569.291	+	2531.678	+	2672.473	+	2529.386	+	2501.057
10	2505.501	-	2666.879	+	2535.403	=	2599.365	+	2519.944	=	2557.738
11	2785.318	+	2899.059	+	2663.504	-	3111.296	+	2669.862	-	2749.009
12	2866.585	+	2903.486	+	2871.114	+	2928.741	+	2863.776	+	2860.399
Win		11		11		7		12		8	

TABLE V: 20-Dimensional comparison results(Part 1).

Func_Num	Mean_PSO	R	Mean_GA	R	Mean_DE	R	Mean_GWO	R	Mean_GO	R	Mean_GPbasedGO
1	52904.89	+	38427.22	+	54598.73	+	30783.65	+	19378.57	=	18987.99
2	791.1697	+	1435.08	+	616.9638	+	531.2367	+	473.6368	=	471.8706
3	636.9382	+	675.4038	+	630.5062	+	614.387	=	609.4576	=	611.4177
4	968.9801	+	949.5482	+	957.3419	+	864.6567	=	932.2348	+	868.1742
5	3389.725	+	3130.209	+	2040.769	+	1514.343	=	1447.686	=	1371.278
6	3.16E+08	+	7.93E+08	+	25570941	+	3841266	+	2683454	+	146310.6
7	2196.214	+	2180.967	+	2225.742	+	2118.254	=	2134.449	+	2095.174
8	2334.213	+	2429.743	+	2317.251	+	2275.405	+	2251.454	+	2244.28
9	2620.435	+	3059.033	+	2519.274	+	2561.716	+	2484.558	+	2474.834
10	5862.466	+	3977.858	=	5745.673	+	4394.218	+	3732.7	=	3577.976
11	4637.618	+	7291.082	+	4381.296	+	4010.602	+	3239.342	=	3267.323
12	3097.133	+	3753.875	+	2900.005	-	3027.249	+	2961.031	+	2915.146
Win		12		11		11		8		6	

TABLE VI: 20-Dimensional comparison results(Part 2).

Func_Num	Mean_ASO	R	Mean_BOA	R	Mean_SA	R	Mean_CS	R	Mean_CRO	R	Mean_GPbasedGO
1	47506.21	+	99751.94	+	56751.55	+	82307.3	+	1.28E+09	+	18987.99
2	704.7262	+	3692.529	+	1069.681	+	5435.199	+	8923.236	+	471.8706
3	644.7519	+	697.827	+	692.8078	+	708.7662	+	738.0695	+	611.4177
4	927.8916	+	1020.596	+	1025.24	+	1064.253	+	1108.797	+	868.1742
5	2416.787	+	5299.146	+	3482.977	+	9737.955	+	14244.68	+	1371.278
6	11609227	+	2.75E+09	+	7888.752	-	3.44E+09	+	8.79E+09	+	146310.6
7	2242.773	+	2301.811	+	2227.635	+	2320.135	+	2535.034	+	2095.174
8	2359.021	+	2909.087	+	2324.389	+	2903.279	+	31644.07	+	2244.28
9	2688.301	+	3675.949	+	2892.854	+	3409.789	+	4881.742	+	2474.834
10	4580.649	=	6886.978	+	4959.543	+	6572.561	+	8394.181	+	3577.976
11	4633.79	+	9374.592	+	7549.17	+	11243.98	+	15367.25	+	3267.323
12	3479.162	+	4029.731	+	3527.145	+	3676.27	+	4493.905	+	2915.146
Win		11		12		11		12		12	

TABLE VII: 20-Dimensional comparison results (Part 3).

Func_Num	Mean_WOO	R	Mean_EMWPA	R	Mean_FNO	R	Mean_CHO	R	Mean_LEE	R	Mean_GPbasedGO
1	43791.76	+	60062.51	+	31885.35	+	35319.07	+	31636.37	+	18987.99
2	580.321	+	606.6398	+	492.6074	+	1268.704	+	481.9139	=	471.8706
3	630.0162	+	658.2631	+	613.9394	+	651.1437	+	607.1674	-	611.4177
4	927.687	+	901.6484	+	887.4723	+	931.0341	+	881.7019	+	868.1742
5	1977.039	+	2829.912	+	1347.751	=	2773.033	+	1185.675	-	1371.278
6	22469984	+	520960	=	19478.4	-	3.73E+08	+	742251.9	+	146310.6
7	2159.721	+	2183.707	+	2118.613	+	2196.076	+	2135.334	+	2095.174
8	2260.329	+	2305.105	+	2239.821	+	2342.964	+	2259.323	+	2244.28
9	2525.473	+	2563.305	+	2497.094	+	2906.536	+	2487.576	+	2474.834
10	3546.754	=	4341.949	+	3132.283	=	3999.009	=	3775.509	=	3577.976
11	4418.292	+	3766.687	+	3113.655	-	6980.613	+	3113.645	-	3267.323
12	2966.472	+	3119.486	+	3011.42	+	3355.557	+	2963.818	+	2915.146
Win		12		12		9		12		9	

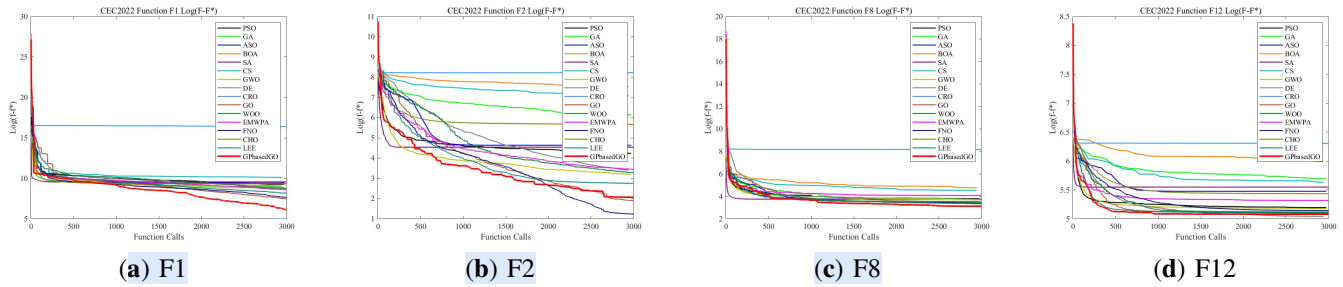


Fig. 5: 10-dimensional convergence plots of the algorithm.

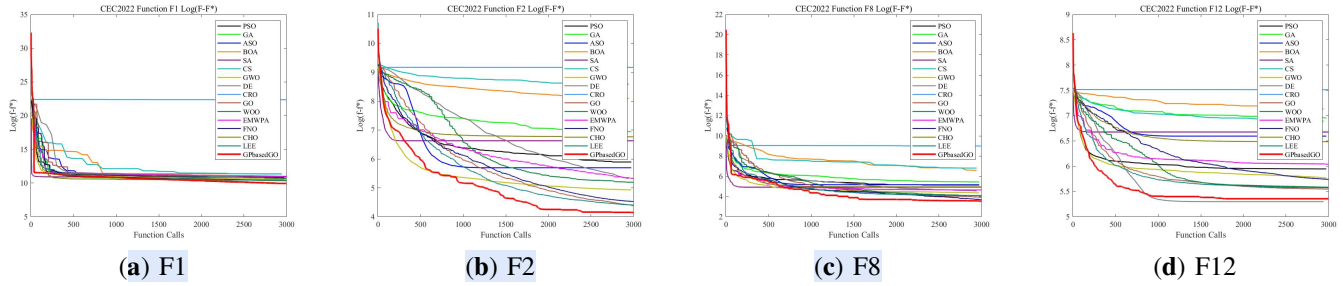


Fig. 6: 20-dimensional convergence plots of the algorithm.

within metaheuristic algorithms. By decoupling the update rule construction from manual design and embedding it within an evolutionary paradigm, the proposed approach addresses key limitations of traditional metaheuristics—namely, empirical dependency, lack of adaptivity, and poor generalizability. The GP-MAs framework not only reduces human intervention and design overhead but also enhances algorithmic robustness by evolving dynamic and problem-specific update mechanisms. To validate the effectiveness of this framework, the Growth Optimizer (GO) algorithm was selected as a case study and enhanced through the integration of GP-driven strategy generation, resulting in the GPbasedGO algorithm. Extensive experiments were conducted on the CEC2022 benchmark suite, where GPbasedGO demonstrated superior performance in terms of convergence accuracy and robustness, particularly in complex and multimodal landscapes. Overall, the proposed GP-MAs framework provides a promising pathway for achieving automated, adaptive, and generalizable optimization in metaheuristic algorithm design.

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